

Project ATLAS — Final Incident Report

SOC Tier 1 Simulation: Multi-Vector Attack Detection, Investigation & Containment

Ivan Yamoah Baoakye

July 5, 2026 | VANPASSSC200 Tenant | Workspace: law-soc-atlas

Executive Summary

Project ATLAS was a five-day self-directed SOC Tier 1 simulation conducted on a personal Microsoft 365 / Azure tenant to replicate the day-to-day responsibilities of a senior Security Operations analyst. The project spanned the full incident lifecycle: infrastructure setup, detection engineering, multi-vector attack simulation, incident investigation, endpoint containment, identity remediation, and proactive threat hunting.

A simulated attacker (Kali Linux ARM64 VM) executed four distinct attack techniques against a Windows 11 victim endpoint (DESKTOP-TRF9U79) and a Microsoft 365 / Entra ID tenant. Microsoft Sentinel, Azure Monitor Agent, Sysmon, Entra ID Protection, and Microsoft Defender's unified XDR portal were used as the detection and response platform. The simulation confirmed that cloud identity-layer detections (anonymous IP, anomalous token) fired correctly, while also surfacing a previously unknown endpoint telemetry gap — a realistic SOC finding with direct operational significance.

Incident Timeline

Date / Time (Local)	Event
27 Jun 2026	Day 2: Detection Engineering & SecurityEvent DCR deployed, custom KQL analytics rules created
3 Jul 2026, 23:14	Day 3: Hydra RDP brute force initiated (T1110.001) & 4625 events generated on DESKTOP-TRF9U79
3 Jul 2026, 23:15	Day 3: Encoded PowerShell executed (T1059.001 + T1027) & whoami output confirmed desktop-TRF9U79
3 Jul 2026, ~23:30	Day 3: Credential harvest phishing simulation launched via Attack Simulation Training & 100% correlation
3 Jul 2026, 23:14 & 4 Jul, 00:07	Day 3: Tor Browser sign-in as soc-tor-test and admin account & 5 anonymous IP detections fired in Microsoft Sentinel
3 Jul 2026, 23:17	Sentinel incident ID 2 created: 'Initial access incident involving one user' & 8/9 alerts, High severity
5 Jul 2026, 10:56	Day 4: Incident triaged, classified True positive & Compromised account
5 Jul 2026, 10:57	Day 4: Endpoint containment & BlockInbound,BlockOutbound applied to DESKTOP-TRF9U79
5 Jul 2026, 11:00	Day 4: SOC Tor Test confirmed compromised in Entra ID Protection
5 Jul 2026, 11:18	Day 4: Threat hunting & SecurityEvent and Sysmon Event tables confirmed 0 records (telemetry gap)
5 Jul 2026, 11:37	Day 4: Conditional Access policy created in Report-only mode

Attack Techniques (MITRE ATT&CK)

Technique ID	Name	Tool / Method	Detection	
T1110.001	Brute Force: Password Guessing	hydra rdp:// module from Kali	4625 events (endpoint); Sentinel rule 4.1	
T1059.001 + T1027	PowerShell + Obfuscation	powershell -enc (Base64 UTF-16)	Sysmon EID 1; Sentinel rule 4.2	
T1566.002	Spearphishing Link	M365 Attack Simulation Training	Attack Simulation Training report	
T1078.004	Valid Accounts: Cloud	soc-tor-test credential use	Entra ID Protection risky sign-in	
T1090.003	Multi-hop Proxy (Tor)	Tor Browser on Mac	Anonymous IP address detection (ML)	

Key Findings

Finding 1: Cloud Identity Detection Confirmed Effective

Entra ID Protection's ML-based Anonymous IP address detection fired automatically within minutes of the Tor Browser sign-in, requiring no custom rule authorship. Five sign-in events were flagged across two Tor exit node IPs (Karlsruhe, DE and Sweden), automatically correlated into a single High-severity incident with 8 alerts. AI-generated technique profiling accurately identified the attack pattern as "token theft and token replay for unauthorized access." The Anomalous Token detection firing alongside the Anonymous IP detection demonstrates defence-in-depth at the identity layer.

Finding 2: Endpoint Telemetry Gap (Critical)

Proactive KQL threat hunting confirmed that both endpoint telemetry pipelines are non-functional: the SecurityEvent table (Windows Security Events via AMA DCR, deployed Day 2) contains 0 records, and the Sysmon Event table (Sysmon via AMA DCR, deployed Day 1) also contains 0 records. As a result, neither the T1110.001 (brute force) nor the T1059.001/T1027 (encoded PowerShell) attacks generated Sentinel alerts through the custom analytics rules created in Day 2. The AMA agent on DESKTOP-TRF9U79 is either not reporting or the DCR data streams are misconfigured. This is a production-grade detection gap requiring immediate investigation.

Recommended remediation: (1) Check AMA agent heartbeat in Azure Monitor (Heartbeat table); (2) verify DCR → resource association in Azure Portal; (3) rerun SecurityEvent | take 10 after confirming agent health; (4) consider deploying Microsoft Defender for Endpoint for richer endpoint telemetry coverage.

Finding 3: Phishing Simulation 100% Effective

The Microsoft Attack Simulation Training credential harvest simulation achieved a 100% compromise rate (1/1 users clicked, 1/1 credentials submitted). This confirms that the tenant has no current phishing-resistant MFA policy or simulated phishing awareness training in place for the tested account. Real-world recommendation: enforce phishing-resistant MFA (FIDO2 / passkeys) and deploy regular Attack Simulation Training campaigns on a monthly cadence.

Response Actions Taken

1. **Incident classified** — Incident ID 2 set to True positive — Compromised account in Microsoft Defender.
 2. **Endpoint contained** — DESKTOP-TRF9U79 isolated via host-based Windows Firewall (BlockInbound,BlockOutbound across all profiles). Containment state confirmed via netsh advfirewall show allprofiles. Firewall restored post confirmation to preserve telemetry pipeline.
 3. **Risky user confirmed compromised** — SOC Tor Test (soc-tor-test@VANPASSSC200.onmicrosoft.com) confirmed compromised in Entra ID Protection. Automated remediation flow initiated by the platform.
 4. **Conditional Access policy deployed** — ATLAS — Block sign-in risk (anonymous IP) created in Report-only mode: targets SOC Tor Test, all cloud apps, blocks on Medium/High sign-in risk. Report-only chosen per best practice (single-admin lab; validate before enforcing).
-

Lab Architecture Summary

The simulation used the following infrastructure, all self-provisioned at zero recurring cost (Azure free trial + M365 Developer tenancy):

- **Attacker:** Kali Linux ARM64 (VirtualBox VM, MacBook Air M4) — Hydra, netcat, base64 PowerShell
 - **Victim endpoint:** Windows 11 ARM64 VM (DESKTOP-TRF9U79, VirtualBox) — Sysmon v15, Azure Monitor Agent, Azure Arc-connected
 - **Network:** VirtualBox NAT; all inter-VM traffic via Mac host (10.0.2.2); RDP port-forwarded host:33389 → guest:3389
 - **SIEM:** Microsoft Sentinel (workspace law-soc-atlas, rg-soc-atlas, East US)
 - **Telemetry:** Sysmon → AMA → Event table; Windows Security Events → AMA → SecurityEvent table (pipeline issue confirmed)
 - **Identity:** Microsoft Entra ID P2 (VANPASSSC200 tenant) — Identity Protection, Conditional Access
 - **Phishing:** Microsoft 365 Defender Attack Simulation Training (Credential Harvest)
-

Lessons Learned

What worked well

- Cloud identity-layer detection (Entra ID Protection) proved robust — ML-based anonymous IP detection required no configuration and fired within minutes

- Microsoft Defender's incident correlation automatically grouped 8 alerts into one actionable incident, reducing triage overhead significantly
- Manual endpoint containment via netsh demonstrated that isolation is achievable without EDR tooling
- Attack Simulation Training provided realistic phishing telemetry without third-party tooling

What to improve

- AMA endpoint telemetry pipeline (SecurityEvent + Sysmon Event) must be validated before relying on custom KQL detection rules — rules that query empty tables produce no alerts
- Clipboard sharing between Mac host and ARM64 VirtualBox VMs is unsupported; future iterations should set up SSH for reliable command delivery
- Entra Conditional Access policy scoping requires disabling Security Defaults first — worth noting for production deployments where Security Defaults may be the only active baseline control
- Hydra's experimental RDP module is blocked by Windows 11 NLA even when NLA is 'disabled' in System Properties — a realistic constraint worth documenting for attack simulation planning

Skills Demonstrated (Job Description Alignment)

JD Requirement	Project ATLAS Evidence
Lead/support incident response incl. investigation, containment, eradication	Triage, contained, eradicated DESKTOP-TRF9U79; confirmed compromised user; created incident report
Analyze logs, alerts, telemetry from security tools and/or cross-systems	Queried SecurityEvent, Event, Sysmon tables; interpreted Entra ID Protection risk scores
Improve detection, response, alerting capabilities	Built 2 custom Sentinel KQL rules (T1110.001, T1059.001+T1027); discovered endpoint telemetry
Improve cloud security posture (Azure/Entra ID)	Deployed Conditional Access risk policy; confirmed compromised identity; audited Defender for Cloud
Balance security best practices with operational requirements	Chased Reports on CA policy to avoid lockout; documented trade-offs explicitly
Govern/enhance SIEM, EDR, endpoint protection tooling	Configured Sentinel DCRs, analytics rules, entity mapping; diagnosed AMA telemetry pipeline
Drive automation and orchestration	Entra ID Protection automated risk scoring; incident auto-correlation in Defender XDR

Project Deliverables

- Day 1: Lab setup — Kali + Windows 11 VMs, Azure Arc onboarding, Sysmon installation, AMA DCR configuration
- Day 2: Detection engineering — SecurityEvent DCR, 2 custom KQL analytics rules, entity mapping
- Day 3: Attack simulation — Hydra, encoded PowerShell, phishing simulation, Tor sign-in
- Day 4: Investigation & containment — incident triage, endpoint isolation, risky user remediation, Conditional Access policy

- Day 5: Final report (this document) + session log PDFs for Days 2–4
- GitHub repository: all session logs, KQL detection queries, architecture diagram, and README (see repository)

Prepared by: Ivan Yamoah Baoakye | **Tenant:** VANPASSSC200.onmicrosoft.com | **Date:** July 5, 2026